STARLIKE COUNTEREXAMPLES TO THE HOT SPOTS CONJECTURE IN EVERY DIMENSION AT LEAST THREE

ABSTRACT. For every integer $n \geq 3$, we construct bounded Lipschitz domains in $\mathbb{R}^n$, starlike about the origin, whose first positive Neumann eigenvalue is simple and admits a real eigenfunction with a unique global maximum at the origin. The domains consist of the unit ball and radial extensions along a periodic frame on the projected faces of a cube. They admit finite semialgebraic descriptions with rational coefficients. The outer radius is 16, and the relative strip width in each cube-face chart is $1/[2(n-1)16^n]$. For every sufficiently fine integer grid the conclusion follows from a uniform extension construction, a directional lower energy estimate, explicit radial trial functions, and eigenfunction convergence both uniformly on the actual domains and in C^2 on smaller balls inside the unit ball. The first eigenpair is identified without constructing the full homogenized angular matrix. The three-dimensional geometry follows the construction of Atar and Burdzy; the proof and the parameter choices are developed for all $n \geq 3$. The grid threshold is existential.

Draft status. This candidate proof was obtained by our team on October 19, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

CONTENTS

1. Introduction 2
1.1. The explicit family and main result 2
1.2. Connection with Atar–Burdzy 3
1.3. Order of the proof 3
2. The domains and their geometry 3
2.1. A finite description 4
2.2. How to read the illustrations 4
2.3. Boundary geometry and averaging 4
3. Uniform extension and compactness 8
3.1. The inner attachment 8
4. A lower bound for the Neumann energies 11
5. The radial problem and explicit spectral separation 14
5.1. The first positive radial eigenfunction 14
5.2. Spherical splitting and the numerical parameters 16
6. Convergence of the first eigenpair 19
7. Uniform and interior convergence 20
7.1. Estimates on the actual domains 20
7.2. Convergence up to the boundary 23
8. The unique interior maximum 23
Effectivity and the dimension restriction 24
References 24

Date: October 20, 2026.
2020 *Mathematics Subject Classification.* Primary 35P05; Secondary 35J25, 35B27.
Key words and phrases. Hot spots conjecture, Neumann eigenvalues, starlike domains, angular perforations, spectral convergence, Sobolev extension.

1. Introduction

For a bounded connected Lipschitz domain $D \subset \mathbb{R}^n$, write

$$0 = \lambda_0(D) < \lambda_1(D) \le \lambda_2(D) \le \cdots$$

for the eigenvalues of the Neumann Laplacian, with multiplicities. The hot spots conjecture asks whether the extrema of a first nonconstant Neumann eigenfunction must occur on the boundary. A particularly strong failure consists of a simple $\lambda_1(D)$ with an associated real eigenfunction having a unique global maximum strictly inside D. We construct such a failure under the additional geometric condition of starlikeness, in each dimension at least three.

1.1. The explicit family and main result.

Let $B_r = \{x \in \mathbb{R}^n : |x| < r\}$, where $|\cdot|$ denotes the Euclidean norm, and define

$$R = 16, \qquad h_n = \frac{1}{2(n-1)16^n}, \qquad \theta_n = 1 - (1-h_n)^{n-1}. \tag{1.1}$$

For $x \neq 0$, set $t(x) = \max_j |x_j|$. Choose any index $i(x)$ attaining this maximum and, for an integer $m \ge 1$, put

$$d_{\mathbb{Z}}(s) = \min_{k\in\mathbb{Z}} |s-k|, \qquad G_{n,m}(x) = \min_{j\neq i(x)} d_{\mathbb{Z}}\left(\frac{mx_j}{t(x)}\right). \tag{1.2}$$

At a tie for the largest absolute coordinate another maximal coordinate contributes $d_{\mathbb{Z}}(\pm m) = 0$; thus $G_{n,m}$ is independent of the choice of $i(x)$. Define

$$\boxed{\Omega_{n,m} = B_1 \cup \left\{x \neq 0 : \ |x| < 16, \quad G_{n,m}(x) < \frac{1}{4(n-1)16^n}\right\}.} \tag{1.3}$$

Theorem 1.1. *For every integer $n \ge 3$, there is an integer $m_0(n)$ such that, for all $m \ge m_0(n)$, the set $\Omega_{n,m}$ in (1.3) is a bounded connected Lipschitz domain, starlike about the origin, with a simple first positive Neumann eigenvalue λ_m. A corresponding real eigenfunction u_m can be normalized by*

$$\int_{\Omega_{n,m}} u_m \, dx = 0, \qquad \int_{\Omega_{n,m}} u_m^2 \, dx = 1, \qquad u_m(0) > 0.$$

It has a continuous representative on $\overline{\Omega_{n,m}}$, and

$$u_m(0) > u_m(x) \qquad \left(x \in \overline{\Omega_{n,m}} \setminus \{0\}\right). \tag{1.4}$$

The eigenfunction is invariant under coordinate permutations and coordinate reflections. In particular,

$$D^2 u_m(0) = -\frac{\lambda_m u_m(0)}{n} I_n < 0. \tag{1.5}$$

Each domain has a finite semialgebraic description with rational coefficients, obtained by removing $2n(2m)^{n-1}$ conical notches from B_{16}.

The values of R and h_n in (1.1) are numerical, and their spectral separation is verified below by explicit inequalities. The integer $m_0(n)$ is not computed. Consequently, a picture made with a convenient coarse grid is a picture of the family, not an individual numerical certificate of (1.4).

The proof gives additional information. With mass density $a_\theta = 1$ in B_1 and $a_\theta = \theta_n$ in $B_{16} \setminus \overline{B}_1$, the first eigenvalues converge to the first positive weighted radial eigenvalue. The consistently signed eigenfunctions converge uniformly on their actual domains to a strictly decreasing radial function $f(|x|)$, and converge with every fixed number of derivatives on smaller balls inside B_1.

1.2. **Connection with Atar–Burdzy.** Atar and Burdzy [1] constructed a three-dimensional starlike Lipschitz counterexample with a simple first positive eigenvalue and a unique maximum at the origin. Their geometry joins a unit-ball core to a fine grid of retained angular directions. Their analysis identifies an effective angular matrix and obtains convergence of the indexed spectrum.

We retain that geometric mechanism, replacing planar angular frames by $(n-1)$-dimensional frames. Our aligned grids simplify the face seams. A directional duality estimate supplies a lower comparison form, and radial trial functions identify the first eigenpair. We do not identify the full effective angular matrix or the higher limiting eigenvalues. The isotropic coefficient in our comparison form is only a lower bound, not a claim that the effective matrix is isotropic.

For comparison of parameters, [1] uses $\theta = (4\pi)^{-1}R^{-5/2}$, with R chosen large. Here the strict radial–angular comparison permits the fixed radius 16 and rational widths (1.1).

1.3. **Order of the proof.** Section 2 gives the finite set description, its geometry, and periodic averaging. Section 3 constructs extensions that preserve the unit ball, including the attachment at radius one. Section 4 establishes the lower energy bound. Section 5 proves the radial ODE facts and the numerical spectral separation. Section 6 identifies the first eigenpair and proves eventual simplicity. Section 7 upgrades convergence to uniform convergence and interior derivative convergence. Section 8 then proves the unique maximum and the Hessian formula.

We use the usual trace, Sobolev, compactness, and fixed-domain extension theorems; see [3, 2]. All estimates required to be uniform in the angular grid are established in the text. Unless specified otherwise, constants may depend on the fixed n, R, h, but never on m.

2. The domains and their geometry

For the geometric and compactness arguments, first fix arbitrary $n \geq 3$, $R > 1$, and $0 < h < 1$. Set

$$d = n-1, \quad Q = (0,1)^d, \quad H = [h/2, 1-h/2]^d, \quad Y_h = Q \setminus H, \quad \theta = |Y_h| = 1-(1-h)^d. \tag{2.1}$$

The frame Y_h is the union of its $2d$ open coordinate boundary strips. Each strip is convex, and strips in different coordinate directions intersect. Since $d \geq 2$, their intersection graph is connected, so Y_h is path connected. Also

$$h \leq \theta \leq dh. \tag{2.2}$$

The lower bound follows from $(1-h)^d \leq 1-h$, and the upper bound from Bernoulli's inequality $(1-h)^d \geq 1-dh$. The integer grid spacing is $\delta = 1/m$. On each face coordinate cube $[-1,1]^d$, there are exactly $(2m)^d$ full period cells.

For $i \in \{1,\dots,n\}$ and $\sigma \in \{-1,1\}$, let $j_1 < \cdots < j_d$ be the remaining indices and define

$$\Psi_{i,\sigma}(y) = \frac{\sigma e_i + \sum_{\ell=1}^{d} y_\ell e_{j_\ell}}{\sqrt{1+|y|^2}}, \qquad y \in [-1,1]^d. \tag{2.3}$$

The image of $\Psi_{i,\sigma}$ is exactly

$$\{\omega \in S^{n-1} : \ \sigma\omega_i \geq |\omega_j| \text{ for every } j \neq i\}.$$

Choosing a maximal absolute coordinate shows that these images cover S^{n-1}; their interiors are disjoint because strict dominance is unique. Their inverses are $y_\ell = \omega_{j_\ell}/|\omega_i|$. Let $U_m \subset S^{n-1}$ be the directions retained by the periodic condition

$$\min_{1\leq \ell \leq d} d_{\mathbb{Z}}(m y_\ell) < h/2. \tag{2.4}$$

All seams are retained, and the tie argument following (1.2) makes this a global definition. Write

$$\Omega_m = B_1 \cup \{r\omega : 0 < r < R, \ \omega \in U_m\}. \tag{2.5}$$

When R, h have the values (1.1), this is $\Omega_{n,m}$.

2.1. **A finite description.** For $a = (a_1, \dots, a_d) \in \{-m, \dots, m-1\}^d$, set

$$K_{i,\sigma,a} = \{x \in \mathbb{R}^n : |x|^2 \geq 1, \quad \sigma x_i \geq 0, \\ (a_\ell + h/2)\sigma x_i \leq m x_{j_\ell} \leq (a_\ell + 1 - h/2)\sigma x_i \quad (1 \leq \ell \leq d)\}. \tag{2.6}$$

Then

$$\Omega_m = B_R \setminus \bigcup_{i,\sigma,a} K_{i,\sigma,a}. \tag{2.7}$$

Indeed, if $x \in K_{i,\sigma,a}$ and $\sigma x_i = 0$, the linear inequalities force all its coordinates to vanish, contradicting $|x|^2 \geq 1$. Hence $\sigma x_i > 0$, and those inequalities imply $|x_{j_\ell}| \leq (1 - h/(2m))\sigma x_i < \sigma x_i$. Thus i is the unique maximal index and all the nondominant ratios lie in central closed intervals of their period cells. Hence the direction of x fails (2.4), including when one of its ratios is an endpoint of a central interval. Since also $|x| \geq 1$, every point of $B_R \cap K_{i,\sigma,a}$ is excluded from Ω_m.

Conversely, let $x \in B_R \setminus \Omega_m$. Then $|x| \geq 1$, and the tie argument following (1.2) shows that there is a unique maximal coordinate i. Put $\sigma = \operatorname{sgn}(x_i)$ and $t = \sigma x_i > 0$. Each ratio $q_\ell = m x_{j_\ell}/t$ lies in $(-m, m)$ and satisfies $d_{\mathbb{Z}}(q_\ell) \geq h/2$. Thus its unique integer part $a_\ell = \lfloor q_\ell \rfloor$ belongs to $\{-m, \dots, m-1\}$, and

$$a_\ell + h/2 \leq q_\ell \leq a_\ell + 1 - h/2.$$

Multiplying by t gives all the inequalities in (2.6), proving (2.7).

The central coordinate boxes are nonempty and pairwise disjoint, and each notch has a unique dominant signed coordinate. Thus there are exactly $2n(2m)^d$ distinct notches. At the numerical values of (1.1), all their defining coefficients are rational. The strict inequality $|x|^2 < 16^2$ defining the outer ball shows that (2.7) is semialgebraic.

2.2. **How to read the illustrations.** The material outside B_1 is a connected union of thin conical sheets. Each excluded notch begins at radius one and opens to the outer boundary. A coordinate hyperplane is entirely retained:

$$\Omega_m \cap \{x_j = 0\} = B_R \cap \{x_j = 0\}, \qquad 1 \leq j \leq n. \tag{2.8}$$

For a nonzero point of this hyperplane a maximal coordinate is nonzero, so a nondominant zero coordinate makes (2.4) hold. Accordingly, the three-dimensional section illustrated below uses an oblique plane.

2.3. **Boundary geometry and averaging.** We record the geometric facts needed for the extension construction, using the notation fixed above. For $a \in \{-m, \dots, m-1\}^d$, write $Q_{a,m} = \delta(a+Q)$ and $H_{a,m} = \delta(a+H)$. The notation Ψ_α denotes any one of the $2n$ cube-face parametrizations used in the definition of U_m.

Proposition 2.1. *The aligned-grid sets Ω_m are bounded, connected Lipschitz domains, starlike about the origin and invariant under every signed coordinate permutation, and*

$$B_1 \subset \Omega_m \subset B_R, \qquad \partial\Omega_m \subset \{1 \leq |x| \leq R\}.$$

There are $a_0, L > 0$ and m_*, depending only on n, R, h, such that, for $m \geq m_*$, every point of $\partial\Omega_m$ has a boundary graph neighborhood of radius a_0/m with Lipschitz constant at most L. Furthermore,

$$\mathbf{1}_{U_m} \rightharpoonup^* \theta \quad \text{in } L^\infty(S^{n-1}). \tag{2.9}$$

Proof. Every closed cube $H_{a,m}$ is inset by $h\delta/2$ from every face of its surrounding period cube. Since ± 1 are grid lines, this also holds at the boundary of each cube-face chart. The maps Ψ_α and their inverses have uniformly bounded first derivatives on the closed face charts. Consequently distinct removed sets on the same face are at spherical distance at least $c_n h\delta$, and each removed set is at distance at least $c_n h\delta$ from the seams of its face. For removed sets on different faces, a shortest path between them must leave the first face and enter the second; the same lower bound

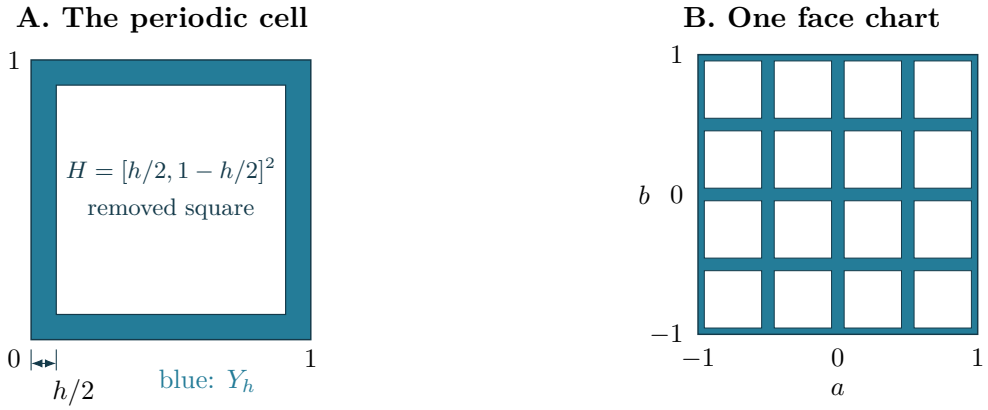


FIGURE 1. The angular frame construction. Left: a fundamental cell of the periodic frame, with the closed square $H = [h/2, 1 - h/2]^2$ removed; opposite cell edges are identified. Right: the 4×4 grid in the chart $(a, b) \in [-1, 1]^2$, with $m = 2$ and $h = 0.18$. Blue directions extend to radius R; white directions stop at radius 1. All six cube-face charts carry this pattern, and their seams belong to the retained frame. These are illustration parameters, not a certified finite-grid counterexample.

follows. The removed sets have diameter comparable to $(1 - h)\delta$. The retained angular set is open because its complement is the finite union of the closed sets $\Psi_\alpha(H_{a,m})$. On each closed face its coordinate image is the union of the grid strips. The strip-intersection argument used for Y_h applies here as well: any transverse grid strip joins all parallel strips. Thus this union is path connected. Every seam is retained, so adjacent face images intersect in the retained set. The face adjacency graph of the cube is connected, and hence U_m is path connected. In particular, the portion $\Omega_m \cap \{1 < |x| < R\}$ is connected, as asserted above.

The radial definition gives openness, boundedness, starlikeness, and the asserted inclusions. It also gives connectedness because every point can be joined to the origin by a radial segment. The membership test is invariant under coordinate permutations and coordinate reflections: $t(x)$ is invariant, the minimum ranges over all nondominant coordinates, and $d_{\mathbb{Z}}$ is even. This gives the asserted symmetries.

Here are the boundary models, including the reentrant corners. In the interior of a face chart use the smooth coordinates

$$X_\alpha(r, y) = r\Psi_\alpha(y).$$

At a corner of a removed angular cube, choose signed coordinates $z_1, \dots, z_k$, $1 \le k \le d$, pointing into its active faces. Away from the two radial ends, the domain has the local model $\bigcup_{j=1}^{k} \{z_j < 0\}$, with the other coordinates unrestricted. At its lower end, put $s = r - 1$; the model is the complement of

$$\{s \ge 0,\ z_1 \ge 0, \dots, z_k \ge 0\}.$$

This includes the case $k = 0$, in which the boundary is just $s = 0$. At the upper end, with $s = R - r$, the model is

$$\{s > 0\} \cap \bigcup_{j=1}^{k} \{z_j < 0\}.$$

If the angular point is in the interior of the retained set, the upper model is simply $\{s > 0\}$. An angular point in the interior of a removed cube gives no boundary point at $r = R$, since it is not in $\overline{\Omega_m}$ there. Thus the upper corner model always has $k \ge 1$.

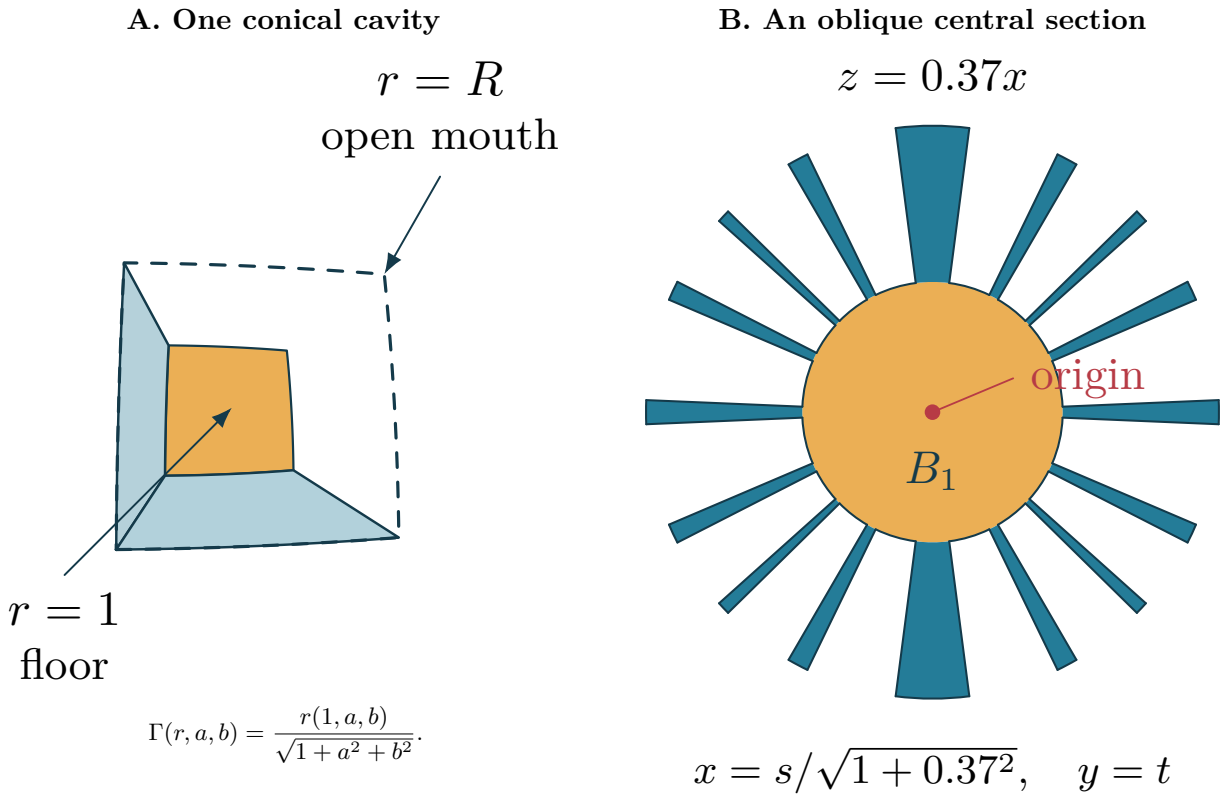


FIGURE 2. Spatial geometry for the illustration values $R = 2.2$, $h = 0.18$, and $m = 2$. Left: the excluded conical region parametrized by $1 < r < R$ and $0.045 < a, b < 0.455$. The gold spherical floor and the blue planar radial walls are parts of the domain boundary. The dashed spherical quadrilateral outlines the mouth of the cavity at $r = R$; its interior is excluded. Two radial walls are omitted to expose the floor. The complete domain consists of connected radial sheets surrounding all such cavities, together with the unit ball. Right: the actual section in the plane $z = 0.37x$, with orthonormal coordinates (s, t). Coordinate-plane sections would instead be full disks of radius R. The red point identifies only the origin. These finite display parameters are not certified as a counterexample.

Choose a_0 sufficiently small compared with h and $1 - h$. The separation just proved and

$$|r\omega - s\eta|^2 = (r - s)^2 + rs|\omega - \eta|^2 \qquad (r, s \geq 1)$$

ensure that a ball of radius $4a_0\delta$ meets at most one removed angular cube and never meets two opposite faces of that cube. For sufficiently large m it also cannot meet both radial ends. If it meets a removed cube, decreasing a_0 once more ensures that the whole ball lies within that cube's face chart, away from the seams. The charts and their inverses have uniform derivative bounds for $1/2 \leq r \leq R + 1$.

We verify the graph assertion directly in Euclidean coordinates. Fix $p \in \partial\Omega_m$ and use the preceding conclusions on $B(p, 4a_0\delta)$. If the ball meets no removed cone, only the smooth outer sphere can be a boundary there. Otherwise discard the angular inequalities whose wall planes miss the ball, since their signs are constant there and keep the ball on their inside-cone sides. The remaining angular inequalities, with their positive sides pointing out of the cone, have the form

$$\ell_\nu(x) = \varepsilon_\nu(x_{j_\nu} - \beta_\nu\sigma x_i) > 0, \qquad \nu \in I \subset \{1, \ldots, d\}, \quad \varepsilon_\nu \in \{-1, 1\}.$$

The two wall parameters in any one coordinate differ by $(1-h)\delta$. The uniform coordinate bounds and the choice of a_0 therefore ensure that at most one of their planes meets the ball. Write $p = X_\alpha(r_p, y_p)$, and choose the coordinate vector v by $v_r = -1$, $v_{y_\nu} = \varepsilon_\nu$ for $\nu \in I$, and $v_{y_\nu} = 0$ otherwise. Set

$$e = \frac{DX_\alpha(r_p, y_p)v}{|DX_\alpha(r_p, y_p)v|}.$$

Every remaining wall plane meets $B(p, 4a_0\delta)$, so $|(y_p)_\nu - \beta_\nu| \le Ca_0\delta$. The formula

$$\ell_\nu(X_\alpha(r, y)) = \frac{\varepsilon_\nu r(y_\nu - \beta_\nu)}{\sqrt{1+|y|^2}}$$

gives

$$D_v(\ell_\nu \circ X_\alpha)(r_p, y_p) = \frac{r_p}{\sqrt{1+|y_p|^2}} + O(a_0\delta).$$

Since $1 \le r_p \le R$, $|y_p|^2 \le d$, and $|DX_\alpha(r_p, y_p)v|$ is bounded above and below uniformly, we obtain $\partial_e \ell_\nu \ge c > 0$ for all these linear walls. For a radial wall $q_\rho(x) = \rho - |x|$, where $\rho = 1$ or R, the same choice gives

$$\partial_e q_\rho(p) = |DX_\alpha(r_p, y_p)v|^{-1} \ge c.$$

Its derivatives vary by $O(a_0\delta)$ in the ball. Hence, for sufficiently large m, $\partial_e q_\rho \ge c/2$ there as well.

Use orthogonal coordinates $x = p + z + te$ with $z \in e^\perp$. Each angular inequality is $t > g_\nu(z)$ for an affine function with uniformly bounded Lipschitz constant. Each radial wall that meets the ball similarly gives $t > g_\rho^{\mathrm{rad}}(z)$: its normal has a uniformly positive component along e. The sphere equation gives this graph on a disk in $e^\perp$ centered at $z = 0$, of fixed radius and with a uniform Lipschitz bound. Since the sphere is at distance $O(a_0\delta)$ from p, that disk contains the projection of $B(p, a_0\delta)$ for all large m. When $I \ne \varnothing$, the local models above give the graph heights

$$\min_{\nu\in I} g_\nu, \qquad \min\{g_1^{\mathrm{rad}}, \min_{\nu\in I} g_\nu\}, \qquad \max\{g_R^{\mathrm{rad}}, \min_{\nu\in I} g_\nu\}$$

for a lateral wall, a lower attachment, and an upper attachment, respectively. In the cases with no angular wall the height is just g_1^{rad} or g_R^{rad}. Finite minima and maxima preserve the uniform Lipschitz bound. Thus $\Omega_m \cap B(p, a_0\delta)$ is exactly the portion of that ball above an L-Lipschitz graph, whose height at $z = 0$ is zero because p is a boundary point. This proves the quantified assertion. For each of the finitely many smaller m, take a sufficiently small neighborhood of p and retain only the walls through p. The error term in the directional calculation then vanishes at p, and the same argument gives a local Lipschitz graph, with no uniform radius needed.

Finally, put $\chi_m(y) = \mathbf{1}_{\{\min_\ell d_{\mathbb{Z}}(my_\ell) < h/2\}}$. Every period cube satisfies $\int_{Q_{a,m}} (\chi_m - \theta)\, dy = 0$. For a continuous b on $[-1,1]^d$, let ω_b be its modulus of continuity and choose $y_a \in \overline{Q_{a,m}}$. Cellwise subtraction gives the quantitative averaging bound

$$\begin{aligned}\left|\int_{[-1,1]^d} b(y)(\chi_m(y) - \theta)\, dy\right| &\le \sum_a \int_{Q_{a,m}} |b(y) - b(y_a)|\, |\chi_m(y) - \theta|\, dy \\ &\le 2^d \omega_b(\sqrt{d}/m) \longrightarrow 0.\end{aligned}$$

Apply this to $b(y) = g(\Psi_\alpha(y))J_\alpha(y)$, with $J_\alpha = \sqrt{\det(D\Psi_\alpha^{\mathsf{T}} D\Psi_\alpha)}$, and sum over the faces for $g \in C(S^{n-1})$. The seams have surface measure zero. Since $|\mathbf{1}_{U_m} - \theta| \le 1$, approximation of $L^1(S^{n-1})$ tests by continuous functions extends this convergence to all such tests and proves (2.9). □

3. Uniform extension and compactness

3.1. **The inner attachment.** The construction below is the dimensional adaptation of the inner attachment extension in Atar–Burdzy [1, Section 6.2]. We give the full argument, including the estimates uniform in the integer grid.

The extension must preserve the trace from the whole unit ball. Applying only an angular extension for $r > 1$ need not do so. We therefore construct a separate extension on a layer of thickness δ at $r = 1$ and join the two extensions there.

Lemma 3.1. *Define the fixed reference domains*

$$D = \big((-1,0) \times Q\big) \cup \big((-1,2) \times Y_h\big), \qquad F = (-1,2) \times Q.$$

There are linear extensions

$$T : H^1(Y_h) \longrightarrow H^1(Q), \qquad S : H^1(D) \longrightarrow H^1(F)$$

that preserve constants and satisfy separate estimates

$$\|Tu\|_{L^2(Q)} \le C_{n,h}\|u\|_{L^2(Y_h)}, \qquad \|\nabla Tu\|_{L^2(Q)} \le C_{n,h}\|\nabla u\|_{L^2(Y_h)}, \tag{3.1}$$

$$\|Su\|_{L^2(F)} \le C_{n,h}\|u\|_{L^2(D)}, \qquad \|\nabla Su\|_{L^2(F)} \le C_{n,h}\|\nabla u\|_{L^2(D)}. \tag{3.2}$$

Both operators equal their input on the original source domain.

Proof. The connectedness of Y_h was proved in Section 2. Thus the connected cylinder $(-1,2) \times Y_h$ meets the connected lower box $(-1,0) \times Q$, proving that D is connected. At the outer rectangular corners, Y_h and D have ordinary orthant models. These are Lipschitz graphs along their inward diagonals, with graph heights given by finite maxima of linear functions. The inner corners of Y_h are complements of orthants; the additional inner boundary models for D are the lower attachment at $t = 0$ and the upper attachment at $t = 2$. These inner models were treated in Proposition 2.1. The gap $h/2$ separates these inner angular boundaries from the outer angular faces of Q, so there are no further corner types.

For completeness, one can obtain the required fixed-domain extensions by the following finite reflection construction. Let G be either Y_h or D and let F_G be the corresponding surrounding rectangle. Choose finitely many nested graph patches $V_j \Subset W_j$ covering ∂G, small enough that reflection of V_j across its graph stays compactly inside W_j. Choose a smooth partition of unity equal to one near $\overline{G}$, with boundary cutoffs supported in the V_j and interior cutoffs compactly supported inside G. In orthogonal coordinates where G lies above $t = \varphi(z)$, reflect the cutoff piece of u by

$$(z,t) \longmapsto (z, 2\varphi(z) - t).$$

This map is bi-Lipschitz, has Jacobian of absolute value one almost everywhere, and has derivative bounded in terms of the graph's Lipschitz constant. Each reflected piece has the same trace from the two sides of the graph and compact support inside W_j. It can therefore be extended by zero outside W_j without a chart-edge jump. Extend the interior pieces by zero as well and sum all the pieces. Sobolev trace gluing gives a function $A_G u \in H^1(F_G)$, and the partition of unity ensures that $A_G u = u$ on G. Change of variables for each reflection, followed by the product rule for the fixed cutoffs, gives

$$\|A_G u\|_{L^2(F_G)} \le C_G\|u\|_{L^2(G)}, \qquad \|\nabla A_G u\|_{L^2(F_G)} \le C_G\big(\|\nabla u\|_{L^2(G)} + \|u\|_{L^2(G)}\big).$$

All cover and cutoff constants are fixed, depending only on n, h.

Write $u_G = |G|^{-1}\int_G u$ and replace this operator by

$$\mathcal{A}_G u = A_G(u - u_G) + u_G \quad \text{on } F_G.$$

It is linear, is still an extension, preserves constants, and is bounded on L^2: Cauchy's inequality gives $|u_G| \le |G|^{-1/2}\|u\|_{L^2(G)}$, so

$$\|\mathcal{A}_G u\|_{L^2(F_G)} \le C_G\|u - u_G\|_{L^2(G)} + |F_G|^{1/2}|u_G| \le C'_G\|u\|_{L^2(G)}.$$

Poincaré's inequality on the connected fixed domain G removes the L^2 term from the gradient estimate:

$$\|\nabla\mathcal{A}_G u\|_{L^2(F_G)} \le C_G\big(\|\nabla u\|_{L^2(G)} + \|u - u_G\|_{L^2(G)}\big) \le C'_G\|\nabla u\|_{L^2(G)}.$$

This fixed-domain Poincaré inequality can itself be seen by contradiction: a sequence of mean-zero functions with unit L^2 norm and gradients tending to zero has, by the established H^1 bound for A_G and Rellich compactness on a rectangle, a strong L^2 limit; that limit is constant on connected G and has mean zero, a contradiction. Taking $\mathcal{A}_G$ for the two choices of G proves the lemma. □

Lemma 3.2. *For fixed $n \ge 3$, $R > 1$, and $0 < h < 1$, and all sufficiently large m, there is a linear operator*

$$E_m : H^1(\Omega_m) \longrightarrow H^1(B_R), \qquad E_m v = v \text{ on } \Omega_m,$$

with $E_m 1 = 1$ and

$$\|E_m v\|_{L^2(B_R)} \le C\|v\|_{L^2(\Omega_m)}, \qquad \|\nabla E_m v\|_{L^2(B_R)} \le C\|\nabla v\|_{L^2(\Omega_m)}. \tag{3.3}$$

Here C depends only on n, R, h, and not on m. In particular there is a uniform Sobolev inequality

$$\|v\|^2_{L^{2n/(n-2)}(\Omega_m)} \le C\left(\|\nabla v\|^2_{L^2(\Omega_m)} + \|v\|^2_{L^2(\Omega_m)}\right). \tag{3.4}$$

Proof. Take m large enough that $\delta < 1/4$ and $1 + 2\delta < R$. Fix a face and a period cube, suppress their indices where convenient, and write

$$\widehat{v}(r, y) = v(X_\alpha(r, y)), \qquad y = \delta(a + z), \quad z \in Q.$$

For $r > 1$ its source in this period is precisely $Y_{a,m} = \delta(a + Y_h)$, whereas for $r < 1$ the entire period is present. For almost every $r \in (1, R)$ apply T to $z \mapsto \widehat{v}(r, \delta(a + z))$ and denote the scaled result by $b_a(r, y)$ on $Q_{a,m}$. Write $T_{a,m}$ for this scaled angular operator. Its separate L^2 bound extends it by density to a bounded map $L^2(Y_{a,m}) \to L^2(Q_{a,m})$, uniformly in m, a. To justify radial differentiation, put $I = (1, R)$. Fubini's theorem gives $\widehat{v} \in H^1(I; L^2(Y_{a,m}))$ and $\widehat{v}(r, \cdot) \in H^1(Y_{a,m})$ for almost every r. For $\eta \in C_c^\infty(I)$, the $L^2(Q_{a,m})$-valued identity

$$\int_I b_a\eta'\,dr = T_{a,m}\left(\int_I \widehat{v}\eta'\,dr\right) = -T_{a,m}\left(\int_I \partial_r\widehat{v}\eta\,dr\right)$$

shows that $\partial_r b_a = T_{a,m}(\partial_r\widehat{v})$. Consequently

$$\begin{aligned}
\int_{Q_{a,m}} |b_a|^2\,dy &\le C\int_{Y_{a,m}} |\widehat{v}|^2\,dy,\\
\int_{Q_{a,m}} |\nabla_y b_a|^2\,dy &\le C\int_{Y_{a,m}} |\nabla_y\widehat{v}|^2\,dy,\\
\int_{Q_{a,m}} |\partial_r b_a|^2\,dy &\le C\int_{Y_{a,m}} |\partial_r\widehat{v}|^2\,dy.
\end{aligned} \tag{3.5}$$

In particular $b_a \in H^1((1, R) \times Q_{a,m})$.

Next introduce the bottom source and target boxes

$$\begin{aligned}
D_a &= \big((1 - \delta, 1) \times Q_{a,m}\big) \cup \big((1 - \delta, 1 + 2\delta) \times Y_{a,m}\big),\\
F_a &= (1 - \delta, 1 + 2\delta) \times Q_{a,m}.
\end{aligned}$$

They are obtained from D, F by the common scaling

$$t = (r - 1)/\delta, \qquad z = y/\delta - a.$$

Apply S under this scaling and call its result s_a. Because all $n = d+1$ coordinates are scaled by the same factor, the L^2 integrals acquire the factor δ^n, and the gradient integrals acquire δ^{n-2}, on both sides of (3.2). Thus

$$\int_{F_a} |s_a|^2 \le C \int_{D_a} |\widehat{v}|^2, \qquad \int_{F_a} |\nabla_{r,y} s_a|^2 \le C \int_{D_a} |\nabla_{r,y} \widehat{v}|^2. \tag{3.6}$$

Choose a fixed smooth $\chi : \mathbb{R} \to [0,1]$ with $\chi = 1$ on $(-\infty, 1]$ and $\chi = 0$ on $[3/2, \infty)$. On the missing portion of this period define

$$(E_m v)(X_\alpha(r,y)) = \begin{cases} \chi((r-1)/\delta) s_a(r,y) + (1 - \chi((r-1)/\delta)) b_a(r,y), & 1 < r < 1+2\delta, \\ b_a(r,y), & 1+2\delta \le r < R. \end{cases} \tag{3.7}$$

On Ω_m set $E_m v = v$.

We verify the gluing before estimating the norm. Both s_a and b_a equal $\widehat{v}$ on their source frame $Y_{a,m}$. In fact, the interpolation in (3.7) is an H^1 function on the entire period box $(1, 1+2\delta) \times Q_{a,m}$ and equals $\widehat{v}$ on its frame. Thus patching it with the original function on the retained part gives exactly this same H^1 function; no internal angular interface remains to be glued. The formula also agrees with the original function in the open retained collars of every lateral period face. By Proposition 2.1, such collars occur on both sides of each face seam as well. Hence neighboring periods and different cube-face charts have matching traces. At $r = 1$, the cutoff selects s_a, and s_a equals $\widehat{v}$ on the whole lower half-box $(1-\delta, 1) \times Q_{a,m}$. Its trace at $r = 1$ is therefore exactly the trace of the unit-ball function, including in the removed angular directions. At the upper end of the interpolation region the cutoff is already zero on a neighborhood, so the formulas agree there as well. There is no further matching condition at $r = R$, because the target is $H^1(B_R)$. There are only finitely many pieces for each m; these trace agreements give a function in $H^1(B_R)$ by Sobolev gluing. The construction is linear and preserves constants.

It remains to estimate the derivative of the cutoff, the only term not immediately controlled by (3.5) and (3.6). Let c_a be the ordinary coordinate mean of $\widehat{v}$ on D_a. The scaled Poincaré inequality on D_a is

$$\int_{D_a} |\widehat{v} - c_a|^2 \le C\delta^2 \int_{D_a} |\nabla_{r,y} \widehat{v}|^2. \tag{3.8}$$

On $J_a = (1+\delta, 1+3\delta/2) \times Q_{a,m}$, the separate L^2 bounds and preservation of constants give

$$\int_{J_a} |s_a - b_a|^2 \le 2 \int_{J_a} |s_a - c_a|^2 + 2 \int_{J_a} |b_a - c_a|^2 \le C \int_{D_a} |\widehat{v} - c_a|^2.$$

The term produced by differentiating the cutoff is $\delta^{-1} \chi'((r-1)/\delta)(s_a - b_a)$. Its support lies in J_a, so (3.8) yields

$$\begin{aligned} \int_{J_a} \left| \delta^{-1} \chi'((r-1)/\delta)(s_a - b_a) \right|^2 &\le C\delta^{-2} \int_{D_a} |\widehat{v} - c_a|^2 \\ &\le C \int_{D_a} |\nabla_{r,y} \widehat{v}|^2. \end{aligned}$$

Thus the cutoff introduces no negative power of δ into the gradient bound.

For $1/2 < r < R$, the maps X_α and their inverses have bounded derivatives, and their Jacobians are bounded above and below by positive constants depending only on n, R. Thus coordinate L^2 and full-gradient integrals are uniformly equivalent to the physical integrals on all source and target boxes. The angular period interiors are disjoint, even between different faces, and so are the bottom boxes D_a in physical coordinates. Each physical source point belongs to at most one angular source and one bottom source; the unchanged portion contributes at most one further copy. Summing the preceding estimates therefore proves (3.3) with a constant independent of m. Applying the ordinary Sobolev inequality on the fixed ball B_R to $E_m v$ yields (3.4). □

In all subsequent limiting statements, $m \to \infty$; the finitely many indices preceding the extension construction are omitted.

Lemma 3.3. *Suppose*

$$\sup_m \left(\|v_m\|_{L^2(\Omega_m)} + \|\nabla v_m\|_{L^2(\Omega_m)} \right) < \infty.$$

After passage to a subsequence there is $v \in H^1(B_R)$ such that $E_m v_m \rightharpoonup v$ in $H^1(B_R)$ and $E_m v_m \to v$ in $L^2(B_R)$. With $a_\theta = 1$ on B_1 and $a_\theta = \theta$ on $B_R \setminus B_1$,

$$\int_{\Omega_m} v_m \longrightarrow \int_{B_R} a_\theta v, \qquad \int_{\Omega_m} v_m^2 \longrightarrow \int_{B_R} a_\theta v^2. \tag{3.9}$$

For two such simultaneously convergent sequences, their inner products converge to the corresponding a_θ-weighted inner product.

Proof. Lemma 3.2 and Rellich compactness on B_R give the subsequence and its stated convergences. For $g \in L^1(B_R)$, polar Fubini gives

$$G(\omega) = \int_1^R g(r\omega) r^{n-1}\, dr \in L^1(S^{n-1}), \qquad \|G\|_{L^1(S^{n-1})} \le \|g\|_{L^1(B_R \setminus B_1)}.$$

The radial definition of Ω_m and (2.9) consequently give

$$\int_{B_R} (\mathbf{1}_{\Omega_m} - a_\theta) g\, dx = \int_{S^{n-1}} (\mathbf{1}_{U_m} - \theta) G\, dS \longrightarrow 0.$$

Thus $\mathbf{1}_{\Omega_m} \rightharpoonup^* a_\theta$ in $L^\infty(B_R)$, with no restriction on the L^1 test.

Write $z_m = E_m v_m$. Strong L^2 convergence gives $z_m \to v$ in $L^1(B_R)$, and Cauchy's inequality gives

$$\|z_m^2 - v^2\|_{L^1(B_R)} \le (\|z_m\|_2 + \|v\|_2)\|z_m - v\|_2 \longrightarrow 0.$$

In each integral over Ω_m, replace v_m by z_m. The resulting integrands converge strongly in L^1, so the preceding weak-star convergence of the indicators proves (3.9). For a second sequence with $q_m = E_m w_m \to w$ in $L^2(B_R)$,

$$\|z_m q_m - vw\|_{L^1(B_R)} \le \|z_m - v\|_2 \|q_m\|_2 + \|v\|_2 \|q_m - w\|_2 \longrightarrow 0.$$

Applying the same indicator limit to this product proves the final assertion. □

4. A lower bound for the Neumann energies

With the density a_θ of Lemma 3.3, set

$$(v, w)_\theta = \int_{B_R} a_\theta v w\, dx, \qquad \|v\|_\theta^2 = (v, v)_\theta.$$

For extensions converging strongly in $L^2(B_R)$, that lemma gives

$$\int_{\Omega_m} v_m\, dx \longrightarrow (v, 1)_\theta, \qquad \int_{\Omega_m} v_m w_m\, dx \longrightarrow (v, w)_\theta. \tag{4.1}$$

On the interior of each cube-face chart, write

$$\chi_m(y) = \mathbf{1}_{\{\min_{1 \le j \le d} d_{\mathbb{Z}}(m y_j) < h/2\}}.$$

The aligned construction makes this the exact retained indicator almost everywhere throughout the face. By the averaging argument in Section 2, $\chi_m \rightharpoonup^* \theta$. The next lemma controls a derivative in a direction along which an indicator is constant.

Lemma 4.1 (A directional lower bound)**.** *Let D be a bounded open rectangular box, let t denote one of its coordinates, and suppose that $0 \le b_m \le 1$ is independent of t. Suppose further that*

$$b_m \rightharpoonup^* \beta \quad \text{in } L^\infty(D), \qquad 0 \le \beta \le 1,$$

where β is constant. Let (z_m) be bounded in $H^1(D)$ and converge strongly in $L^2(D)$ to z. If $A \in C^1(\overline{D})$ is strictly positive on $\overline{D}$, then

$$\liminf_{m\to\infty} \int_D A b_m |\partial_t z_m|^2 \ge \beta \int_D A |\partial_t z|^2. \tag{4.2}$$

Proof. Weak compactness and the strong L^2 limit imply $z \in H^1(D)$. After permuting coordinates, write $D = I \times D'$ and $b_m(t,y) = \widetilde{b}_m(y)$ almost everywhere. For almost every y, Sobolev slicing gives $z_m(\cdot, y) \in H^1(I)$. Integrate by parts on this slice against $(A\phi)(\cdot, y) \in C_c^1(I)$, multiply by $\widetilde{b}_m(y)$, and apply Fubini. No derivative of the measurable coefficient is required. Thus, for $\phi \in C_c^\infty(D)$,

$$\begin{aligned}\int_D A b_m \partial_t z_m \, \phi &= -\int_D b_m z_m \, \partial_t(A\phi) \\ &\longrightarrow -\beta \int_D z \, \partial_t(A\phi) = \beta \int_D A \partial_t z \, \phi.\end{aligned}$$

Here the part containing $z_m - z$ tends to zero by strong L^2 convergence, and the remaining part converges by weak-star convergence against the L^1 function $z\partial_t(A\phi)$. Completing the square gives

$$\int_D A b_m |\partial_t z_m|^2 \ge 2 \int_D A b_m \partial_t z_m \, \phi - \int_D A b_m \phi^2.$$

After passage to the lower limit, the right side is

$$2\beta \int_D A \partial_t z \, \phi - \beta \int_D A \phi^2.$$

This expression equals

$$\beta \|\partial_t z\|^2_{L^2(D, A\,dx)} - \beta \|\partial_t z - \phi\|^2_{L^2(D, A\,dx)}.$$

Since A is bounded above and below by positive constants, $C_c^\infty(D)$ is dense in this weighted L^2 space. Taking the supremum over ϕ proves the assertion, including the case $\beta = 0$. □

Proposition 4.2 (Lower comparison form)**.** *Suppose that $v_m \in H^1(\Omega_m)$ satisfies*

$$\sup_m \left(\|v_m\|^2_{L^2(\Omega_m)} + \int_{\Omega_m} |\nabla v_m|^2 \, dx \right) < \infty.$$

After passage to a subsequence, the extensions in Lemma 3.2 converge weakly in $H^1(B_R)$ and strongly in $L^2(B_R)$ to a function v. For every such subsequence,

$$\liminf_{m\to\infty} \int_{\Omega_m} |\nabla v_m|^2 \, dx \ge \mathcal{Q}(v), \tag{4.3}$$

where

$$\begin{aligned}\mathcal{Q}(v) = &\int_{B_1} |\nabla v|^2 \, dx \\ &+ \theta \int_1^R \int_{S^{n-1}} \left(r^{n-1} |\partial_r v|^2 + \frac{r^{n-3}}{n(n-1)} |\nabla_{S^{n-1}} v|^2 \right) dS \, dr.\end{aligned} \tag{4.4}$$

Proof. Lemma 3.3 supplies the asserted subsequence. Fix any subsequence with the stated convergences and retain the index m. It remains to prove the energy inequality.

Fix one face chart $\Psi = \Psi_\alpha$ from (2.3) and set $s = |y|^2$, $F = D\Psi$, $G = F^T F$, and $J = \sqrt{\det G}$. Direct differentiation gives

$$\begin{aligned} G &= (1+s)^{-1} I - (1+s)^{-2} y y^T, \\ G^{-1} &= (1+s)(I + y y^T), \qquad J = (1+s)^{-n/2}. \end{aligned} \tag{4.5}$$

For $y \neq 0$, G has eigenvalue $(1+s)^{-1}$ on $y^\perp$ and $(1+s)^{-2}$ along y. Thus $\det G = (1+s)^{-(d+1)} = (1+s)^{-n}$, which gives the displayed Jacobian. The smallest and largest eigenvalues of G^{-1} are $1+s$ and $(1+s)^2$, respectively, since $d \geq 2$. At $y = 0$, both matrices are the identity and the same formulas hold.

Let $X(r, y) = r\Psi(y)$, and write

$$z_m = (E_m v_m) \circ X, \qquad z = v \circ X.$$

The uniform coordinate bounds from Section 2 imply that (z_m) is bounded in $H^1(C)$ and converges strongly in $L^2(C)$ to z, where $C = (1, R) \times (-1, 1)^d$. Since $|\Psi| = 1$ and $F^{\mathsf{T}}\Psi = 0$,

$$(DX)^{\mathsf{T}} DX = \begin{pmatrix} 1 & 0 \\ 0 & r^2 G \end{pmatrix}, \qquad |\det DX| = r^{n-1} J.$$

The angular gradient differentiates $\omega \mapsto v(r\omega)$ on the unit sphere. The chain rule gives

$$\nabla_{S^{n-1}} v(r, \Psi(y)) = F G^{-1} \nabla_y z, \qquad |\nabla_{S^{n-1}} v(r, \Psi(y))|^2 = \nabla_y z^{\mathsf{T}} G^{-1} \nabla_y z.$$

The metric identities hold first for smooth functions and extend to H^1 by density and the bi-Lipschitz coordinate change. Because $E_m v_m = v_m$ and their weak gradients agree on Ω_m, the exact energy $\mathcal{E}_{\alpha,m}$ on the retained part of this face is

$$\int_1^R \int_{(-1,1)^d} J \chi_m \left(r^{n-1} |\partial_r z_m|^2 + r^{n-3} \nabla_y z_m^T G^{-1} \nabla_y z_m \right) dy\, dr. \tag{4.6}$$

For each $k \in \{1, \ldots, d\}$ choose $j(k) \neq k$; this is possible precisely because $d \geq 2$. For example, use the cyclic permutation. Define the subslab indicator

$$b_{k,m}(y) = \mathbf{1}_{\{d_{\mathbb{Z}}(m y_{j(k)}) < h/2\}}.$$

It satisfies $b_{k,m} \leq \chi_m$ and is independent of y_k. Its defining one-dimensional periodic strip indicator has mean h; the cellwise averaging argument of Section 2 therefore gives $b_{k,m} \rightharpoonup^* h$. By Fubini, this convergence and $\chi_m \rightharpoonup^* \theta$ hold on the whole cylinder C. The pointwise inequalities

$$\begin{aligned} \chi_m \nabla_y z_m^T G^{-1} \nabla_y z_m &\geq (1+s) \chi_m \sum_{k=1}^d |\partial_{y_k} z_m|^2 \\ &\geq (1+s) \sum_{k=1}^d b_{k,m} |\partial_{y_k} z_m|^2 \end{aligned} \tag{4.7}$$

involve no double counting: the kth component of the gradient occurs once, with a coefficient at most χ_m. Overlap between the slabs therefore causes no loss.

Apply Lemma 4.1 first with $t = r$, $b_m = \chi_m$, $\beta = \theta$, and $A = J r^{n-1}$. Apply it next, separately for every k, with $t = y_k$, $b_m = b_{k,m}$, $\beta = h$, and $A = J r^{n-3}(1+s)$. Both weights are strictly positive and C^1 on $\overline{C}$. The radial term and all selected coordinate terms are nonnegative, so the lower limit of their finite sum is at least the sum of their lower limits. Consequently the lower limit of (4.6) is at least

$$\int_1^R \int_{(-1,1)^d} J \left(\theta r^{n-1} |\partial_r z|^2 + h r^{n-3} (1+s) |\nabla_y z|^2 \right) dy\, dr.$$

Since $s \le d = n-1$, the largest-eigenvalue bound in (4.5) gives

$$(1+s)|\nabla_y z|^2 \ge \frac{1}{1+s}\nabla_y z^T G^{-1}\nabla_y z \ge \frac{1}{n}|\nabla_{S^{n-1}} v|^2.$$

By (2.2), $h/n \ge \theta/[n(n-1)]$.

The face interiors are disjoint, and the omitted seams and the interface $r=1$ have zero volume. Thus the total energy decomposes as

$$\int_{\Omega_m} |\nabla v_m|^2\,dx = \int_{B_1} |\nabla E_m v_m|^2\,dx + \sum_{\alpha=1}^{2n} \mathcal{E}_{\alpha,m}.$$

Weak $H^1(B_R)$ convergence gives

$$\liminf_m \int_{B_1} |\nabla E_m v_m|^2\,dx \ge \int_{B_1} |\nabla v|^2\,dx.$$

Apply the finite-sum lower-limit inequality once more and insert the face bounds above. Their surface integrals cover S^{n-1} up to a null set, yielding exactly (4.3). □

5. The radial problem and explicit spectral separation

5.1. **The first positive radial eigenfunction.** For the fixed R, h, θ, put

$$p(r) = \begin{cases} r^{n-1}, & 0 < r < 1, \\ \theta r^{n-1}, & 1 < r < R. \end{cases} \tag{5.1}$$

Let $\mathcal{V}$ be the space of measurable radial profiles for which $U(x) = f(|x|)$ belongs to $H^1(B_R)$, identifying profiles that agree almost everywhere. Each such profile lies in $H^1(a,R)$ for every $0 < a < R$. With f' denoting its local weak derivative, polar coordinates give

$$\|U\|^2_{H^1(B_R)} = |S^{n-1}| \int_0^R r^{n-1}\big(f^2 + |f'|^2\big)\,dr.$$

These facts follow by smooth approximation and rotational averaging of the approximants. The radial subspace is closed in $H^1(B_R)$, being the common fixed space of the continuous rotation operators. Define

$$\lambda_{\mathrm{rad}} = \inf_{\substack{f \in \mathcal{V}\setminus\{0\} \\ \int_0^R pf\,dr = 0}} \frac{\int_0^R p(f')^2\,dr}{\int_0^R pf^2\,dr}. \tag{5.2}$$

The admissible class is nonempty: subtract the p-weighted mean from r^2, for example. The weights are comparable to the ordinary radial weights because $0 < \theta < 1$ is fixed. For a weighted mean-zero function $U = f(|x|)$, put $c = |B_R|^{-1}\int_{B_R} U\,dx$. Expanding the square and then using the ordinary Poincaré inequality on the fixed ball gives

$$\begin{aligned} \|U\|^2_\theta &\le \int_{B_R} a_\theta |U-c|^2\,dx \le \int_{B_R} |U-c|^2\,dx \\ &\le C_P \int_{B_R} |\nabla U|^2\,dx \le \frac{C_P}{\theta}\int_{B_R} a_\theta |\nabla U|^2\,dx. \end{aligned}$$

Here C_P depends only on n, R. Thus $\lambda_{\mathrm{rad}} \ge \theta/C_P > 0$. For attainment, normalize a minimizing sequence by $\int_0^R pf_k^2\,dr = 1$. The associated radial functions are bounded in $H^1(B_R)$. Weak compactness and Rellich give a weak H^1 and strong L^2 limit in the closed radial subspace. The weighted mean and norm pass to the limit, and weak lower semicontinuity of $\int a_\theta|\nabla U|^2$ proves attainment.

Lemma 5.1. *The eigenvalue λ_{rad} is simple in the radial space. Its eigenfunction can be signed and normalized so that*

$$f(0) > 0, \qquad |S^{n-1}| \int_0^R p(r) f(r)^2 \, dr = 1.$$

It is continuous on $[0, R]$, smooth on $[0, 1)$ as a radial function in the ball and on $(1, R]$, and satisfies

$$-(pf')' = \lambda_{\mathrm{rad}} pf, \qquad f'(0) = f'(R) = 0, \qquad f(1-) = f(1+), \quad f'(1-) = \theta f'(1+). \tag{5.3}$$

It has exactly one zero in $(0, R)$ and is strictly decreasing. Moreover,

$$p(r) f'(r) = -\lambda_{\mathrm{rad}} \int_0^r p(s) f(s) \, ds < 0 \qquad (0 < r < R), \tag{5.4}$$

with one-sided derivatives at the interface, and

$$D^2 \big[f(|x|) \big]_{x=0} = -\frac{\lambda_{\mathrm{rad}} f(0)}{n} I_n. \tag{5.5}$$

Proof. The Euler equation for a minimizer is

$$\int_0^R p f' \varphi' \, dr = \lambda_{\mathrm{rad}} \int_0^R p f \varphi \, dr \qquad (\varphi \in \mathcal{V}). \tag{5.6}$$

Differentiating the Rayleigh quotient gives this identity for mean-zero test profiles. For arbitrary φ, apply it to

$$\varphi_0 = \varphi - \frac{\int_0^R p\varphi \, dr}{\int_0^R p \, dr}.$$

The derivative is unchanged, and $\int_0^R pf \, dr = 0$ removes the constant on the right. For each $a \in (0, 1)$, the profile belongs to $H^1(a, R)$, so it is continuous across one and up to R. Equation (5.6) says in distributions that $q' = -\lambda_{\mathrm{rad}} pf$ on (a, R), where $q = pf'$. Both q and q' lie in $L^2(a, R)$, so $q \in H^1(a, R)$ has an absolutely continuous representative and is continuous through one. Integration by parts with test profiles supported away from zero but having arbitrary trace at R gives $q(R) = 0$. This proves the flux transmission and Neumann conditions. The ODE then gives smoothness on either side of the interface.

To justify the equation at the origin as well, let $\psi \in C_c^\infty(B_1)$ and set

$$\overline{\psi}(r) = \frac{1}{|S^{n-1}|} \int_{S^{n-1}} \psi(r\omega) \, dS.$$

This is the profile of the smooth rotational average of ψ, which vanishes near $r = 1$. Its extension by zero for $r \geq 1$ therefore belongs to $\mathcal{V}$. Radiality and (5.6) give

$$\int_{B_1} \nabla[f(|x|)] \cdot \nabla\psi \, dx = |S^{n-1}| \int_0^1 r^{n-1} f' \overline{\psi}' \, dr = \lambda_{\mathrm{rad}} \int_{B_1} f(|x|) \psi \, dx.$$

Thus $f(|x|)$ solves $-\Delta f = \lambda_{\mathrm{rad}} f$ weakly on the whole ball, with no point singularity at zero. Interior regularity makes it smooth at zero. Radial smoothness implies $f'(0) = 0$. For a fixed eigenvalue the regular radial solution is uniquely determined by its value at zero. Indeed, let g be the difference of two regular solutions with the same value there. Smoothness gives $r^{n-1} g'(r) \to 0$ at zero, so integration of the ODE gives

$$r^{n-1} g'(r) = -\lambda_{\mathrm{rad}} \int_0^r s^{n-1} g(s) \, ds.$$

If $M_a = \max_{0 \leq r \leq a} |g(r)|$, a second integration, now using $g(0) = 0$, yields

$$|g(r)| \leq \frac{\lambda_{\mathrm{rad}} r^2}{2n} M_a, \qquad M_a \leq \frac{\lambda_{\mathrm{rad}} a^2}{2n} M_a.$$

Choose a small enough that the factor is less than one. Then $g = 0$ on $[0, a]$. Ordinary ODE uniqueness propagates this identity to one and, through the two interface conditions, to R. Thus evaluation at zero is injective on the radial eigenspace. Since it is a linear map into $\mathbb{R}$, the eigenspace is one-dimensional, and every nonzero eigenfunction has $f(0) \neq 0$. Choose its sign so that $f(0) > 0$ and rescale it to satisfy $|S^{n-1}| \int_0^R pf^2\, dr = 1$.

We give a variational proof of the nodal assertion. Mean zero forces a sign change. On either side of one, the first-order system $f' = q/p$, $q' = -\lambda_{\mathrm{rad}} pf$, together with continuity of f, q at one, shows that zero value and zero flux at any positive radius would force the zero solution. In particular, $f(R) \neq 0$ because $q(R) = 0$. At every zero in $(0, R)$, the continuous flux is nonzero; hence $f' = q/p$ has a fixed nonzero sign in a neighborhood of that zero, including across one. Thus each zero is isolated, and at an interface zero both one-sided derivatives are nonzero. Since neither endpoint is a zero, compactness of $[0, R]$ implies that there are only finitely many zeros.

Suppose there were at least three nodal intervals. Let f_1, f_2 be the restrictions of f to two of them, extended by zero. At every interior endpoint of a nodal interval the trace of f vanishes, so the weak derivative of f_j is $\mathbf{1}_{I_j} f'$, where I_j is its nodal interval. At zero or R no extension beyond the profile interval is required. Thus both restrictions belong to $\mathcal{V}$, and testing (5.6) with either restriction gives

$$\int_0^R p(f_j')^2\, dr = \lambda_{\mathrm{rad}} \int_0^R pf_j^2\, dr, \qquad j = 1, 2.$$

Each mean $M_j = \int_0^R pf_j\, dr$ is nonzero, since f_j has a fixed nonzero sign on its nodal interval. Thus $M_2 f_1 - M_1 f_2$ is nonzero and has weighted mean zero. The disjoint supports of f_1, f_2 show that this combination has quotient λ_{rad}. It is another minimizer, hence an eigenfunction proportional to f by simplicity. That is impossible because the combination vanishes on the third nodal interval. Thus there are exactly two nodal intervals and a single zero z. After the prescribed sign choice, $f > 0$ before z and $f < 0$ after z.

Set $J(r) = \int_0^r p(s)f(s)\, ds$. For $0 < r \leq z$, this is positive. For $z < r < R$, mean zero gives $J(r) = -\int_r^R p(s)f(s)\, ds > 0$. Integration of the ODE from zero gives (5.4), using $pf' \to 0$ at zero and flux continuity at one. Hence f is strictly decreasing. Finally, radial symmetry makes the Hessian of $f(|x|)$ at zero a scalar matrix. Its trace is $-\lambda_{\mathrm{rad}} f(0)$ by the equation, which proves (5.5). □

5.2. **Spherical splitting and the numerical parameters.**

Lemma 5.2 (Spherical splitting)**.** *For $v \in H^1(B_R)$, define*

$$\overline{v}(r) = \frac{1}{|S^{n-1}|} \int_{S^{n-1}} v(r, \omega)\, dS(\omega), \qquad v^\perp = v - \overline{v}.$$

Then both parts belong to $H^1(B_R)$, and

$$\|v\|_\theta^2 = \|\overline{v}\|_\theta^2 + \|v^\perp\|_\theta^2, \qquad \mathcal{Q}(v) = \mathcal{Q}(\overline{v}) + \mathcal{Q}(v^\perp).$$

If $(v, 1)_\theta = 0$, then

$$\mathcal{Q}(v) \geq \lambda_{\mathrm{rad}} \|\overline{v}\|_\theta^2 + \frac{1}{nR^2} \|v^\perp\|_\theta^2. \tag{5.7}$$

Proof. Rotations act isometrically on $H^1(B_R)$, so their average, which is spherical averaging, is bounded on that space. On every annulus $a < r < R$, spherical averaging is also a bounded linear functional on the angular L^2 space. The weak differentiation argument used in Lemma 3.2 therefore gives

$$(\overline{v})'(r) = \frac{1}{|S^{n-1}|} \int_{S^{n-1}} \partial_r v(r, \omega)\, dS(\omega) \quad \text{for almost every } r \in (0, R).$$

Consequently $v^\perp$ and $\partial_r v^\perp$ have zero spherical mean at almost every radius. These identities eliminate the norm and radial-energy cross terms; the angular derivative of $\overline{v}$ vanishes. This proves the orthogonal splittings. It also shows that $(\overline{v}, 1)_\theta = (v, 1)_\theta = 0$, so the radial Rayleigh principle gives the first term of (5.7).

For the other term, use the spherical Poincaré inequality at almost every radius:

$$\int_{S^{n-1}} |\nabla_{S^{n-1}} v^\perp|^2 \, dS \geq (n-1) \int_{S^{n-1}} |v^\perp|^2 \, dS.$$

Dropping the nonnegative radial energy in (4.4) gives

$$\begin{aligned} \mathcal{Q}(v^\perp) &\geq (n-1) \int_0^1 r^{n-3} \int_{S^{n-1}} |v^\perp|^2 \, dS \, dr \\ &\quad + \frac{\theta}{n} \int_1^R r^{n-3} \int_{S^{n-1}} |v^\perp|^2 \, dS \, dr \\ &\geq \frac{1}{nR^2} \|v^\perp\|_\theta^2. \end{aligned}$$

Here $r^{-2} \geq R^{-2}$ on $(0, R)$ and $n - 1 \geq 1/n$. Combining this bound with the radial term proves the assertion. □

Proposition 5.3 (An explicit choice in every dimension)**.** *For every integer $n \geq 3$, set*

$$R = 16, \qquad h = h_n = \frac{1}{2(n-1)16^n}, \qquad \theta = 1 - (1 - h_n)^{n-1}. \tag{5.8}$$

Then

$$0 < \lambda_{\mathrm{rad}} < \frac{1}{256n} = \frac{1}{nR^2}. \tag{5.9}$$

Proof. Positivity was established in the variational construction of (5.2). Applying (2.2) gives

$$0 < \theta \leq (n-1)h_n = \frac{1}{2 \cdot 16^n}. \tag{5.10}$$

Both trial profiles below are continuous, constant near zero, and piecewise smooth with bounded derivative. Their radial functions are therefore Lipschitz and belong to $\mathcal{V}$. The Rayleigh form domain imposes no endpoint Neumann condition on a trial profile.

First suppose $n \geq 4$. Use the radial trial function

$$g(r) = \begin{cases} 1, & 0 \leq r \leq 1, \\ \dfrac{r^{2-n} - 2^{2-n}}{1 - 2^{2-n}}, & 1 < r < 2, \\ 0, & 2 \leq r \leq 16. \end{cases}$$

Let $d\mu = p(r)\, dr$, $M = \mu([0, 16])$, and $b = M^{-1} \int g \, d\mu$. Subtracting b makes the trial function admissible and does not change its energy. It is nonzero because g equals one on the core and zero on $[2, 16]$, both of positive μ-measure. On $(1, 2)$,

$$g'(r) = -\frac{(n-2)r^{1-n}}{1 - 2^{2-n}}.$$

Direct integration gives

$$E := \int_0^{16} p|g'|^2 \, dr = \frac{\theta(n-2)}{1 - 2^{2-n}}, \qquad M = \frac{1 + \theta(16^n - 1)}{n} \leq \frac{3}{2n}.$$

Since $\mu([0,1]) = 1/n$ and $\mu([2,16]) = \theta(16^n - 2^n)/n$, the pairwise variance identity yields

$$D := \int (g-b)^2 \, d\mu = \frac{1}{2M} \iint (g(r) - g(s))^2 \, d\mu(r) \, d\mu(s) \geq \frac{\mu([0,1])\mu([2,16])}{M} \geq \frac{2\theta(16^n - 2^n)}{3n}.$$

The first inequality retains both ordered core–outer-shell contributions, on which the squared difference is exactly one. It follows that

$$\lambda_{\mathrm{rad}} \leq \frac{E}{D} \leq \frac{3n(n-2)}{2(1-2^{2-n})(16^n - 2^n)} \leq \frac{4n(n-2)}{16^n}.$$

For the last inequality, observe that

$$(1-2^{2-n})(1-8^{-n}) \geq \frac{3}{4}\left(1 - \frac{1}{4096}\right) > \frac{3}{8}.$$

Finally,

$$4n^2(n-2) < 16^{n-2} \qquad (n \geq 4).$$

Indeed, it holds at $n = 4$, where $128 < 256$, and the ratio of the successive polynomial factors satisfies

$$\frac{(n+1)^2(n-1)}{n^2(n-2)} = \left(1 + \frac{1}{n}\right)^2 \left(1 + \frac{1}{n-2}\right) \leq \frac{25}{16}\frac{3}{2} = \frac{75}{32} < 16 \qquad (n \geq 4).$$

Induction therefore proves the displayed inequality for every $n \geq 4$. This proves (5.9) in these dimensions.

For $n = 3$, (5.8) gives $h = 1/16384$ and $\theta \leq 1/8192$ by (5.10). Use instead

$$g(r) = \begin{cases} 1, & 0 \leq r \leq 1, \\ 1/r, & 1 < r \leq R. \end{cases}$$

For general $R > 1$, its mass, first moment, second moment, and energy are

$$M = \frac{1 + \theta(R^3 - 1)}{3}, \quad A_1 = \frac{1}{3} + \frac{\theta(R^2 - 1)}{2},$$
$$A_2 = \frac{1}{3} + \theta(R - 1), \qquad E = \theta\left(1 - \frac{1}{R}\right).$$

The function is nonconstant on sets of positive weighted measure. Subtracting its weighted mean A_1/M therefore gives a nonzero admissible trial. Expanding the moments gives

$$MA_2 - A_1^2 = \frac{\theta(R-1)^3}{9} + \frac{\theta^2(R-1)^4}{12}.$$

Hence its denominator is

$$A_2 - \frac{A_1^2}{M} = \frac{\theta(R-1)^3[4 + 3\theta(R-1)]}{12[1 + \theta(R^3 - 1)]}.$$

Therefore

$$\lambda_{\mathrm{rad}} \leq \frac{12[1 + \theta(R^3 - 1)]}{R(R-1)^2[4 + 3\theta(R-1)]}.$$

For fixed $R > 1$, the expression on the right increases with θ: its derivative has the sign of $4(R^3 - 1) - 3(R - 1) = (R-1)(2R+1)^2 > 0$. At $R = 16$ and $\theta = 1/8192$ it is

$$\frac{12287}{9843900} < \frac{1}{768},$$

because $12287 \cdot 768 = 9436416 < 9843900$. This proves (5.9) also for $n = 3$. □

6. Convergence of the first eigenpair

Let $f = f(r)$ be the first radial weighted eigenfunction supplied by Lemma 5.1, normalized by

$$\|f\|_\theta = 1, \qquad (f,1)_\theta = 0, \qquad f(0) > 0.$$

In the following proposition, f also denotes the function $x \mapsto f(|x|)$ on B_R. Let λ_m be the first positive Neumann eigenvalue of Ω_m.

Proposition 6.1 (Convergence and simplicity of the first eigenpair)**.** *For the parameters in Proposition 5.3,*

$$\lambda_m \longrightarrow \lambda_{\mathrm{rad}}.$$

The eigenvalue λ_m is simple for all sufficiently large m. If the corresponding eigenfunction is normalized and oriented by

$$\int_{\Omega_m} u_m \, dx = 0, \qquad \int_{\Omega_m} u_m^2 \, dx = 1, \qquad \int_{\Omega_m} u_m f \, dx > 0,$$

then

$$E_m u_m \longrightarrow f \quad \text{strongly in } L^2(B_R), \qquad \|u_m - f\|_{L^2(\Omega_m)} \longrightarrow 0. \tag{6.1}$$

Proof. Since $f \in H^1(B_R)$, its restriction to Ω_m belongs to $H^1(\Omega_m)$. Subtract its actual mean

$$c_m = \frac{1}{|\Omega_m|} \int_{\Omega_m} f \, dx.$$

The bulk indicator convergence established in Lemma 3.3, tested against $1, f, f^2$, gives

$$|\Omega_m| \longrightarrow (1,1)_\theta > 0, \qquad \int_{\Omega_m} f \, dx \longrightarrow 0, \qquad \int_{\Omega_m} f^2 \, dx \longrightarrow 1.$$

Consequently $c_m \to 0$ and

$$\int_{\Omega_m} (f - c_m)^2 \, dx = \int_{\Omega_m} f^2 \, dx - \frac{\left(\int_{\Omega_m} f \, dx\right)^2}{|\Omega_m|} \longrightarrow 1.$$

Thus $f - c_m$ is a nonzero mean-zero Rayleigh trial for all large m. The same indicator limit, tested against $|\nabla f|^2$, gives

$$\int_{\Omega_m} |\nabla f|^2 \, dx \longrightarrow \int_{B_R} a_\theta |\nabla f|^2 \, dx = \lambda_{\mathrm{rad}}.$$

The final equality follows by testing (5.6) with f and using $\|f\|_\theta = 1$. All four fixed tests above lie in $L^1(B_R)$. The Neumann Rayleigh principle therefore yields

$$\limsup_{m \to \infty} \lambda_m \le \lambda_{\mathrm{rad}}. \tag{6.2}$$

Every positive-eigenvalue Neumann eigenfunction has mean zero by testing its weak equation with 1. For an L^2-normalized first eigenfunction, testing with itself also gives $\int_{\Omega_m} |\nabla u_m|^2 \, dx = \lambda_m$. Start from any subsequence of indices tending to infinity and any choice of real normalized first eigenfunctions u_m on it. The upper bound (6.2) gives a uniform energy bound for all large m. Passing further, we may assume $\lambda_m \to \ell \ge 0$, and Lemma 3.3 gives

$$E_m u_m \rightharpoonup v \quad \text{in } H^1(B_R), \qquad E_m u_m \to v \quad \text{in } L^2(B_R).$$

Equation (4.1) implies

$$\|v\|_\theta = 1, \qquad (v,1)_\theta = 0.$$

Proposition 4.2 and Lemma 5.2 now give

$$\begin{aligned}\lambda_{\mathrm{rad}} &\geq \ell \geq \mathcal{Q}(v)\\ &\geq \lambda_{\mathrm{rad}}\|\overline{v}\|_\theta^2 + \frac{1}{nR^2}\|v^\perp\|_\theta^2\\ &= \lambda_{\mathrm{rad}} + \left(\frac{1}{nR^2} - \lambda_{\mathrm{rad}}\right)\|v^\perp\|_\theta^2.\end{aligned}$$

The coefficient in parentheses is strictly positive by Proposition 5.3. Consequently $v^\perp = 0$ and every inequality in the chain is an equality: $\ell = \mathcal{Q}(v) = \lambda_{\mathrm{rad}}$. For radial v, the form $\mathcal{Q}(v)$ is its weighted radial Dirichlet energy. Together with its unit weighted norm and zero weighted mean, this shows that v attains (5.2). The simplicity assertion in Lemma 5.1 gives $v = f$ or $v = -f$. Since the initial index subsequence and the eigenfunction choices were arbitrary, every such sequence has a further subsequence whose extension limit is f or $-f$, and whose eigenvalue limit is λ_{rad}. This proves convergence of the full eigenvalue sequence.

We now obtain simplicity and the sign choice from a single pairing argument. Whenever a subsequence satisfies $E_m u_m \to \varepsilon f$ in $L^2(B_R)$ with $\varepsilon \in \{-1, 1\}$, the fixed-test limit above gives

$$\begin{aligned}\left|\int_{\Omega_m} u_m f\,dx - \varepsilon\right| &\leq \|E_m u_m - \varepsilon f\|_{L^2(B_R)}\|f\|_{L^2(B_R)}\\ &\quad + \left|\int_{\Omega_m} f^2\,dx - 1\right| \longrightarrow 0.\end{aligned}$$

Suppose that for infinitely many m the first eigenspace contained a nonzero real eigenfunction with $\int_{\Omega_m} u_m f\,dx = 0$. Normalize one such function at each index and pass to a subsequence. The preceding classification gives an extension limit εf, while the displayed pairing limit gives $0 = \varepsilon$, a contradiction. Consequently, for all sufficiently large m, pairing with f defines an injective linear map from the real first eigenspace into $\mathbb{R}$. That nonzero eigenspace must therefore be one-dimensional. This proves eventual simplicity, and the positive pairing selects a unique sign after L^2 normalization.

For this sign choice, the pairing limit excludes $-f$ as a cluster limit, so every convergent subsequence of extensions has limit f. If the full sequence failed to converge to f in $L^2(B_R)$, a subsequence staying a positive distance from f would, by Lemma 3.3, have a further subsequence converging to f, a contradiction. Finally,

$$\|u_m - f\|_{L^2(\Omega_m)} \leq \|E_m u_m - f\|_{L^2(B_R)} \longrightarrow 0,$$

which completes (6.1). □

7. Uniform and interior convergence

7.1. **Estimates on the actual domains.** Throughout this section, m is above the fixed threshold required by Lemma 3.2. The next estimates use only the uniform Sobolev inequality (3.4). In particular, their constants do not depend on the sizes of individual boundary charts.

Lemma 7.1. *There are $C > 0$ and $s_0 > 0$, independent of m, such that the following holds. Let $x \in \overline{\Omega_m}$, $0 < s \leq s_0$, and let $v \in H^1(\Omega_m \cap B(x, 2s))$ satisfy $-\Delta v = F$ with zero Neumann data on the physical boundary, where $F \in L^\infty$. More precisely, require*

$$\int_{\Omega_m \cap B(x,2s)} \nabla v \cdot \nabla \varphi\,dy = \int_{\Omega_m \cap B(x,2s)} F\varphi\,dy$$

for every H^1 test function whose support stays away from $\partial B(x, 2s)$; no zero trace is required on $\partial\Omega_m$. For every constant $c \in \mathbb{R}$,

$$\|v - c\|_{L^\infty(\Omega_m \cap B(x,s))} \leq C\left[s^{-n/2}\|v - c\|_{L^2(\Omega_m \cap B(x,2s))} + s^2\|F\|_\infty\right]. \tag{7.1}$$

Also, for every fixed $\Lambda < \infty$, a Neumann eigenfunction with $0 \le \lambda \le \Lambda$ satisfies

$$\|u\|_{L^\infty(\Omega_m)} \le C_\Lambda \|u\|_{L^2(\Omega_m)}. \tag{7.2}$$

Proof. Write $w = v - c$, take $k = s^2\|F\|_\infty + \varepsilon$ with $\varepsilon > 0$, and set $U = |w| + k$. We give the truncated tests explicitly, so the iteration does not presuppose higher integrability. For $p \ge 2$ and $L > k$, put

$$U_L = \min\{U, L\}, \qquad V_L = U\, U_L^{p/2-1}.$$

Let η be an ambient smooth cutoff compactly supported in $B(x, 2s)$. The function $\eta^2 w U_L^{p-2}$ is an admissible H^1 test, because its dependence on w is Lipschitz for fixed L, p, k. The derivative of wU_L^{p-2} has coefficient at least U_L^{p-2}, and almost everywhere

$$|\nabla V_L| \le \frac{p}{2} U_L^{p/2-1} |\nabla w|, \qquad |w|^2 U_L^{p-2} \le V_L^2, \qquad |w| U_L^{p-2} \le k^{-1} V_L^2.$$

Set $\mathcal{I}_L = \int \eta^2 U_L^{p-2} |\nabla w|^2\, dy$, with integrals over $\Omega_m \cap B(x, 2s)$. The weak equation and Young's inequality, absorbing half of $\mathcal{I}_L$ from the cutoff cross term, give

$$\mathcal{I}_L \le 4 \int |\nabla \eta|^2 V_L^2\, dy + \frac{2\|F\|_\infty}{k} \int \eta^2 V_L^2\, dy.$$

Using the displayed bound for ∇V_L and $\|F\|_\infty / k \le s^{-2}$ now yields

$$\int_{\Omega_m} |\nabla(\eta V_L)|^2\, dy \le C p^2 \int_{\Omega_m} (|\nabla \eta|^2 + s^{-2}\eta^2) V_L^2\, dy. \tag{7.3}$$

The cutoff vanishes in a neighborhood of the artificial sphere $\partial B(x, 2s)$. Thus extending ηV_L by zero to the rest of Ω_m introduces no jump and gives a function in $H^1(\Omega_m)$. Consequently (3.4) applies directly to it.

Put $\chi = n/(n-2)$. If $s \le r < t \le 2s$, choose a cutoff equal to one on $B(x, r)$, compactly supported in $B(x, t)$, and with gradient bounded by $2/(t - r)$. For $s \le s_0 \le 1$, the Sobolev inequality and (7.3), followed by $V_L^2 \le U^p$, imply

$$\|V_L\|^2_{L^{2\chi}(\Omega_m \cap B(x,r))} \le \frac{Cp^2}{(t-r)^2} \int_{\Omega_m \cap B(x,t)} U^p\, dy.$$

Whenever the integral on the right is finite, monotone convergence as $L \to \infty$ gives

$$\|U\|_{L^{p\chi}(\Omega_m \cap B(x,r))} \le \left[\frac{Cp^2}{(t-r)^2}\right]^{1/p} \|U\|_{L^p(\Omega_m \cap B(x,t))}.$$

The right side is finite initially at $p = 2$, and this estimate supplies the next integrability exponent at every step. Iterate with $p_j = 2\chi^j$ and radii $r_j = s + s2^{-j}$, decreasing from $2s$ to s. Here $r_j - r_{j+1} = s2^{-j-1}$, so the accumulated factor is

$$\prod_{j \ge 0} \left[\frac{C p_j^2 4^{j+1}}{s^2}\right]^{1/p_j} \le C s^{-n/2}.$$

Indeed, $\sum_j 1/p_j = n/4$, $\log p_j = O(j)$, and $\sum_j j/p_j < \infty$. Consequently

$$\|U\|_{L^\infty(\Omega_m \cap B(x,s))} \le C s^{-n/2} \|U\|_{L^2(\Omega_m \cap B(x,2s))}.$$

Since $|\Omega_m \cap B(x, 2s)| \le |B(x, 2s)|$, the constant part k contributes at most Ck. Letting $\varepsilon \downarrow 0$ proves (7.1).

For (7.2), use the same truncations with $w = u$, $k = \varepsilon > 0$, and $\eta = 1$, but test the global eigenfunction equation directly. The test uU_L^{p-2} is admissible, and its right side is integrable because $u^2U_L^{p-2} \le L^{p-2}u^2$. The same coercivity and chain-rule estimates give

$$\int_{\Omega_m} U_L^{p-2}|\nabla u|^2\,dx \le \lambda \int_{\Omega_m} u^2U_L^{p-2}\,dx \le \lambda \int_{\Omega_m} V_L^2\,dx,$$
$$\int_{\Omega_m} |\nabla V_L|^2\,dx \le \frac{p^2\lambda}{4}\int_{\Omega_m} V_L^2\,dx.$$

Thus no boundedness of λu is assumed. Apply (3.4) and let $L \to \infty$ at each stage where $U \in L^p$. Monotone convergence gives

$$\|U\|_{p\chi} \le [C(1+p^2\Lambda)]^{1/p}\|U\|_p.$$

The same exponents p_j and summability as above give $\|U\|_\infty \le C_\Lambda\|U\|_2$, uniformly in m, ε. Finally, $\|U\|_2 \le \|u\|_2 + \varepsilon|B_R|^{1/2}$; letting $\varepsilon \downarrow 0$ proves (7.2). □

Lemma 7.2. *Every Neumann eigenfunction on each Ω_m has a continuous representative on $\overline{\Omega_m}$. For the indices covered by Lemma 7.1, (7.1) also controls its boundary values.*

Proof. For the sufficiently large indices under consideration, (7.2) makes $F = \lambda u$ bounded. The same conclusion holds for each of the finitely many omitted indices: its fixed Lipschitz domain has its own Sobolev inequality, and the global truncated iteration above applies with a constant depending on that domain and eigenvalue. Thus it suffices, for every fixed m, to consider bounded forcing on the Lipschitz domain Ω_m. Interior continuity is classical. At a boundary point write the domain locally as $x_n > \varphi(x')$ after a rigid change of coordinates, with φ Lipschitz. The flattening map $T(y', y_n) = (y', y_n + \varphi(y'))$ is bi-Lipschitz and has Jacobian determinant one. Shrink the flattened neighborhood to a box $C = (-a, a)^n$, and set $C_+ = C \cap \{y_n > 0\}$, $\widetilde{u} = u \circ T$, and $\widetilde{F} = F \circ T$. For every $\psi \in H^1(C_+)$ supported away from the lateral and upper faces, compose with T^{-1} and extend by zero outside the chart inside Ω_m. This is an admissible physical H^1 test, with arbitrary trace on the physical boundary. The bi-Lipschitz Sobolev chain rule and change of variables give

$$\int_{C_+} A(y)\nabla\widetilde{u}\cdot\nabla\psi\,dy = \int_{C_+}\widetilde{F}\psi\,dy, \qquad A(y) = [DT(y)]^{-1}[DT(y)]^{-\mathsf{T}},$$

with no restriction on the trace at $y_n = 0$. The coefficient matrix is bounded and uniformly elliptic on this fixed chart. Reflect $\widetilde{u}$ and $\widetilde{F}$ evenly across $y_n = 0$, obtaining $u_{\mathrm{e}} \in H^1(C)$ and $F_{\mathrm{e}} \in L^\infty(C)$. With $J = \mathrm{diag}(1, \dots, 1, -1)$, set $A_{\mathrm{e}} = A$ above the interface and

$$A_{\mathrm{e}}(y', -t) = JA(y', t)J \qquad (t > 0).$$

For $\psi \in C_c^\infty(C)$, put $\psi_+(y', t) = \psi(y', t) + \psi(y', -t)$ on C_+. It is an admissible upper-half test, including its generally nonzero trace on $t = 0$. Change of variables in the lower half gives

$$\int_C A_{\mathrm{e}}\nabla u_{\mathrm{e}}\cdot\nabla\psi\,dy = \int_{C_+} A\nabla\widetilde{u}\cdot\nabla\psi_+\,dy$$
$$= \int_{C_+}\widetilde{F}\psi_+\,dy = \int_C F_{\mathrm{e}}\psi\,dy.$$

This proves the weak equation across the flattened boundary. The reflected matrix remains bounded and uniformly elliptic, and $F_{\mathrm{e}} \in L^q(C)$ for every finite $q > n/2$. Interior De Giorgi–Nash–Moser regularity for divergence equations with measurable coefficients and this forcing gives a locally Hölder continuous representative; see [2, Chapter 8]. Transforming back proves continuity up to this boundary point. The local representatives agree almost everywhere in the interior and hence everywhere there by continuity. Density of Ω_m in its closure extends this agreement to boundary

overlaps. They therefore define a representative on the whole closure. Its supremum on each relative open neighborhood equals its essential supremum: a larger continuous value would persist on a neighborhood meeting the open domain in a set of positive volume. Thus (7.1), in its stated range of indices, also holds pointwise up to the physical boundary. No uniform Hölder constant in m is asserted or used. □

7.2. **Convergence up to the boundary.** For the remainder of the proof take R, h as in (1.1), and use the consistent sign from Proposition 6.1. Write $u_0(x) = f(|x|)$, with f from Lemma 5.1.

Proposition 7.3. *For every fixed $0 < \rho < 1$ and integer $k \geq 0$,*

$$\sup_{x \in \overline{\Omega_m}} |u_m(x) - u_0(x)| \longrightarrow 0, \qquad \|u_m - u_0\|_{C^k(\overline{B}_\rho)} \longrightarrow 0. \tag{7.4}$$

Proof. By (6.1), put $\eta_m = \|u_m - u_0\|_{L^2(\Omega_m)} \to 0$. The eigenvalues are bounded, so (7.2) gives $\|u_m\|_\infty \leq C$ and $\|\lambda_m u_m\|_\infty \leq C$. Define the modulus

$$\omega_0(t) = \sup\{|u_0(x) - u_0(y)| : x, y \in \overline{B}_R,\ |x - y| \leq t\}.$$

It tends to zero because u_0 is continuous on the compact ball. Fix $s \in (0, s_0]$ and $x \in \overline{\Omega_m}$. Apply (7.1) to $v = u_m$ and the constant $c = u_0(x)$. On the larger intersection with $B(x, 2s)$,

$$\|u_m - u_0(x)\|_{L^2(\Omega_m \cap B(x,2s))} \leq \eta_m + C s^{n/2} \omega_0(2s).$$

Consequently,

$$\sup_{x \in \overline{\Omega_m}} |u_m(x) - u_0(x)| \leq C\big[s^{-n/2}\eta_m + \omega_0(2s) + s^2\big]. \tag{7.5}$$

First let $m \to \infty$ with s fixed and then let $s \downarrow 0$. This proves uniform convergence. Here $c = u_0(x)$ is constant in the running variable, so the local weak equation retains the same forcing $\lambda_m u_m$.

For the derivative assertion set $w_m = u_m - u_0$. On the fixed ball B_1,

$$-\Delta w_m = \lambda_m w_m + (\lambda_m - \lambda_{\mathrm{rad}}) u_0. \tag{7.6}$$

For fixed $0 < a < b < 1$ and integer $j \geq 0$, interior estimates for the Laplacian, the uniform eigenvalue bound, and (7.6) give

$$\|w_m\|_{H^{j+2}(B_a)} \leq C_{a,b,j} \left(\|w_m\|_{H^j(B_b)} + |\lambda_m - \lambda_{\mathrm{rad}}| \, \|u_0\|_{H^j(B_b)} \right).$$

The constants are independent of m. For any desired Sobolev order, choose finitely many nested balls between B_ρ and B_1, and iterate this estimate with $j = 0, 2, 4, \ldots$. The starting L^2 convergence follows from $B_1 \subset \Omega_m$; the eigenvalue difference tends to zero by Proposition 6.1, and u_0 is smooth on B_1. Thus $w_m \to 0$ in every $H^q(B_\rho)$. Choose an integer $q > k + n/2$ and apply Sobolev embedding to obtain the stated $C^k(\overline{B}_\rho)$ convergence. □

8. The unique interior maximum

Proof of Theorem 1.1. The finite description and geometric conclusions follow from (2.6)–(2.7) and Proposition 2.1. Proposition 6.1 gives simplicity for all sufficiently large m; use its normalized and consistently signed eigenfunctions u_m. Lemma 7.2 gives continuous representatives on $\overline{\Omega_m}$.

By Proposition 7.3, $u_m(0) \to f(0) > 0$. Thus $u_m(0) > 0$ for all sufficiently large m, as required by the normalization in the theorem. Let S be any signed coordinate permutation. The domain is invariant under S. Since S is orthogonal, a change of variables in the weak Neumann equation shows that $u_m \circ S$ lies in the same one-dimensional eigenspace. Its L^2 norm is unchanged, hence $u_m \circ S = \pm u_m$. Continuity makes this identity pointwise on $\overline{\Omega_m}$. Evaluation at zero, where $S0 = 0$ and $u_m(0) > 0$, fixes the plus sign. The eigenfunction is smooth in B_1, so we may differentiate these identities at zero. Reflection symmetry gives $\nabla u_m(0) = 0$ and vanishing off-diagonal Hessian entries; permutation symmetry makes the diagonal entries equal. Evaluating $-\Delta u_m = \lambda_m u_m$ at zero proves (1.5), whose scalar coefficient is negative because $\lambda_m > 0$ and $u_m(0) > 0$.

We next obtain a neighborhood that works for all sufficiently large m. By (5.5), choose $0 < \rho < 1$ and $a > 0$ such that

$$D^2 u_0(x) \le -2aI_n \qquad (|x| \le \rho).$$

Interior C^2 convergence gives $D^2 u_m \le -aI_n$ on the same ball for all large m. Since $\nabla u_m(0) = 0$, Taylor's integral formula along the segment from zero to x gives

$$\begin{aligned} u_m(x) - u_m(0) &= \int_0^1 (1-t) x^{\mathsf{T}} D^2 u_m(tx) x \, dt \\ &\le -\frac{a}{2}|x|^2 \qquad (0 < |x| \le \rho). \end{aligned} \tag{8.1}$$

Write $\varepsilon_m = \sup_{\overline{\Omega_m}} |u_m - u_0| \to 0$. For every $x \in \overline{\Omega_m}$, uniform convergence gives

$$u_m(0) - u_m(x) \ge f(0) - f(|x|) - 2\varepsilon_m. \tag{8.2}$$

Strict radial decrease provides the fixed number $H_\rho = f(0) - f(\rho) > 0$. For all sufficiently large m, $\varepsilon_m < H_\rho/3$, so whenever $|x| \ge \rho$,

$$u_m(0) - u_m(x) \ge H_\rho - 2\varepsilon_m > H_\rho/3 > 0.$$

Together with (8.1), this proves (1.4). All thresholds depend only on the fixed dimension and its fixed parameters, so one integer $m_0(n)$ satisfies every conclusion. □

Corollary 8.1. *For all sufficiently large m, the eigenfunction normalized as in Theorem 1.1 satisfies*

$$u_m(0) - \max_{\partial\Omega_m} u_m \ge \frac{f(0) - f(1)}{2} > 0.$$

Proof. The boundary maximum exists by compactness and continuity. Since $B_1 \subset \Omega_m$, every boundary point has radius at least one. Thus (8.2) and the monotonicity of f give

$$u_m(0) - \max_{\partial\Omega_m} u_m \ge f(0) - f(1) - 2\varepsilon_m.$$

The result follows once $\varepsilon_m \le (f(0) - f(1))/4$. □

Effectivity and the dimension restriction. Extracting a numerical bound for $m_0(n)$ from this proof would require quantitative bounds for eventual simplicity and for the uniform and interior C^2 convergence used above.

There are two structural uses of $d = n - 1 \ge 2$: the cell frame Y_h is connected, and each angular derivative is controlled on a retained slab defined by a different angular coordinate. When $d = 1$, the cell frame is the disjoint union $(0, h/2) \cup (1 - h/2, 1)$, and no different coordinate is available for the slab argument in Proposition 4.2. Thus this argument gives no planar counterexample.

References

[1] R. Atar and K. Burdzy, *A starlike 3-dimensional counterexample to the hot spots conjecture*, arXiv:2610.19908, version 1 (2026). https://arxiv.org/abs/2610.19908.

[2] D. Gilbarg and N. S. Trudinger, *Elliptic Partial Differential Equations of Second Order*, Classics in Mathematics, Springer-Verlag, Berlin, 2001. https://doi.org/10.1007/978-3-642-61798-0.

[3] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton Mathematical Series, vol. 30, Princeton University Press, Princeton, NJ, 1970.